INNOVA JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATION
in preparation for General Certificate of Education Advanced Level
Higher 1

CANDIDATE NAME

CLASS

INDEX NUMBER

CHEMISTRY 8873/01

Paper 1 Multiple Choice

14 Sep 2018

1 hour

Additional Materials: Data Booklet
Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

Write your index number, name and class on all the work you hand in.
Write in soft pencil.
Do not use staples, paper clips, highlighters, glue or correction fluid.

There are **thirty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A**, **B**, **C** and **D**.
Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

1	**B**	6	**D**	11	**A**	16	**B**	21	**C**	26	**C**
2	**A**	7	**A**	12	**D**	17	**A**	22	**B**	27	**D**
3	**D**	8	**B**	13	**C**	18	**C**	23	**D**	28	**C**
4	**C**	9	**B**	14	**D**	19	**B**	24	**C**	29	**D**
5	**B**	10	**C**	15	**D**	20	**B**	25	**B**	30	**D**

This document consists of **12** printed pages.

For each question there are four possible answers, **A, B, C,** and **D**. Choose the **one** you consider to be correct.

1 A giant molecule contains a large amount of carbon, mainly of isotopes ^{12}C and ^{13}C. It was found that the relative atomic mass of carbon in the molecule is 12.20.

What is the ratio of ^{12}C and ^{13}C?

A 3:1 **B** 4:1 **C** 3:4 **D** 1:4

Answer: B

Let the fraction of ^{12}C to be x and the fraction of ^{13}C to be 1-x

$12x + 13(1-x) = 12.2$

$12x + 13 – 13x = 12.2$

$x = 0.8$

$^{12}C : {}^{13}C = 0.8 : 0.2 = 4 : 1$

2 Which statements about relative molecular mass are correct?

1 It is the sum of the relative atomic masses of all the atoms within the molecule.

2 It is the ratio of the average mass of a molecule to the mass of a ^{12}C atom.

3 It is the ratio of the mass of 1 mol of molecules to the mass of 1 mol of ^{1}H atoms.

4 It is the ratio of the mass of 1 mol of molecules to the mass of 1 mol of ^{12}C atoms.

A 1 only

B 1 and 3 only

C 2 and 3 only

D 1, 2 and 4 only

Answer: A

Option 2 is wrong. It is the ratio of the average mass of a molecule to 1/12 of the mass of a ^{12}C atom.

Option 3 and 4 are wrong. It is the ratio of the mass of 1 mol of molecules to the mass of 1/12 of 1 mol of ^{12}C atoms.

3 *Use of the Data Booklet is relevant to this question.*

Some isotopes are unstable and decompose naturally. In one type of decomposition, a neutron in the nucleus decomposes to form a proton, which is retained in the nucleus, and an electron, which is expelled from the atom.

Which change describes a process of this sort?

A $^{11}C \longrightarrow {}^{12}C$

B $^{22}Na \longrightarrow {}^{22}Ne$

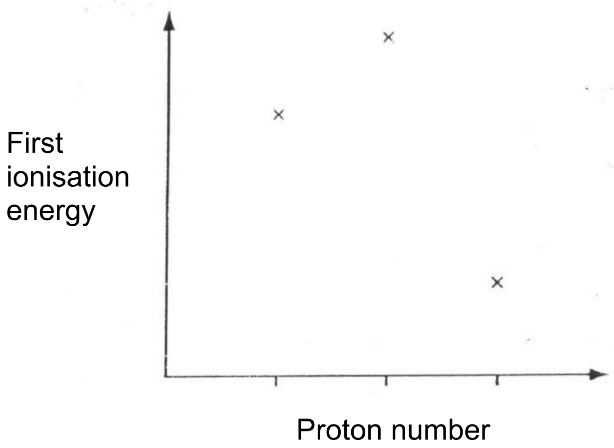
C $^{32}P \longrightarrow {}^{31}P$

D $^{40}K \longrightarrow {}^{40}Ca$

Answer: D

When a neutron in the nucleus decomposes to form a proton, mass number remains the same but proton number increases by 1.

4 Three successive elements in the Periodic Table have first ionisation energies which have the pattern shown in the diagram.



What could be the first element of this sequence?

A N

B O

C F

D Na

Answer: C

Option A is wrong. If N is the first element, the second element will be O. However, there is a decrease in first I.E. between N and O element as there is a pair of electrons in the same p orbital in O which experience inter-electronic repulsion.

Option B is wrong. If O is the first element, the second and third element is F and He which first I.E. increases from O to F to Ne.

Option D is wrong. If Na is the first element, the second element is Mg and Al. There is a slight decrease in I.E. between Mg and Al which is not as shown by the diagram above (large decrease in first I.E. between second and third element.

5 Which of the following ions contains half-filled d-orbitals?

1 Mn^{2+}

2 Cu^{+}

3 Fe^{2+}

4 Fe^{3+}

A 1 and 2 only

B 1 and 4 only

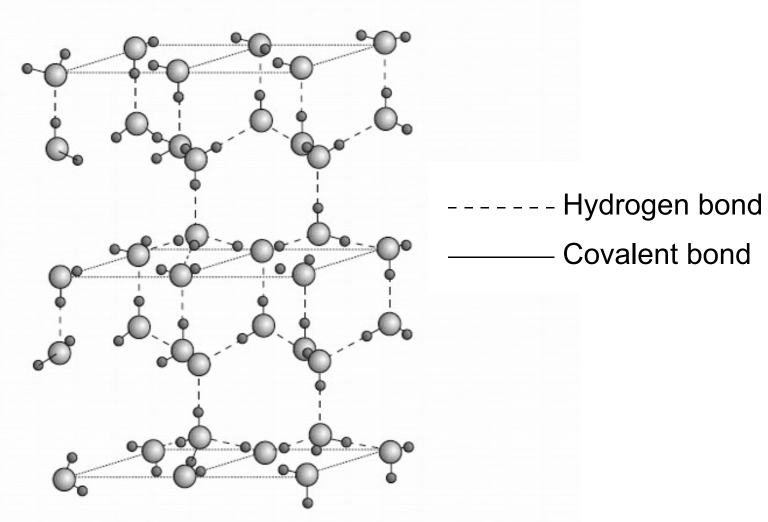
C 2 and 3 only

D 3 and 4 only

Answer: B

The valence electronic configuration of the following ions:
Mn^{2+}: $3d^5$
Cu^{+}: $3d^{10}$
Fe^{2+}: $3d^6$
Fe^{3+}: $3d^5$

6 Ice is the crystalline form of water. The diagram below shows part of the structure of ice.



Which of the following statements is **not** true about ice?

A Ice has a lower density than water at 0 °C due to its open structure.

B The bond angle about oxygen in ice is 109.5°.

C Ice does not conduct electricity.

D Ice has a giant covalent structure.

Answer: D

Ice has simple molecular structure.

7 Which of the following pairs of substances **does not** include a giant structure and a simple molecular structure?

A aluminum and silicon(IV) oxide

B aluminium oxide and aluminium chloride

C silicon and chlorine

D silicon and silicon(IV) chloride

Answer: A

Al_2O_3 has giant ionic structure and SiO_2 has giant covalent structure.
Al_2O_3 has giant ionic structure and $AlCl_3$ has simple molecular structure.
Si has giant covalent structure and Cl_2 has simple molecular structure.
Si has giant covalent structure and $SiCl_4$ has simple molecular structure.

8 The reduction of ethanal to ethanol using $H_2(g)$ is shown in the equation below.

$$\underset{\text{ethanal}}{CH_3CHO(l)} + H_2(g) \rightarrow \underset{\text{ethanol}}{CH_3CH_2OH(l)}$$

substance	ΔH_c^θ / kJ mol^{-1}
ethanal	–1170
hydrogen	–286
ethanol	–1370

Given the standard enthalpy change of combustion in the table above, what is the enthalpy change of the reduction of ethanal to ethanol?

A +86 kJ mol^{-1} **B** –86 kJ mol^{-1} **C** –2826 kJ mol^{-1} **D** +2826 kJ mol^{-1}

Answer: B

$\Delta H_r^\theta = \sum n\Delta H_c^\theta$ (reactants) $- \sum m\Delta H_c^\theta$ (products)
$= [-1170+(-286)] - (-1370) = -86$ kJ mol^{-1}

9 The bond dissociation energy of H–F is 565 kJ mol^{-1}. Which equation correctly describes the reaction whereby 565 kJ of energy is released?

A $HF(l) \rightarrow HF(g)$

B $H(g) + F(g) \rightarrow HF(g)$

C $HF(g) \rightarrow H(g) + F(g)$

D $\frac{1}{2} H_2(g) + \frac{1}{2} F_2(g) \rightarrow HF(g)$

Answer: B

Bond dissociation energy: $HF(g) \rightarrow H(g) + F(g)$ +565 kJ mol^{-1}

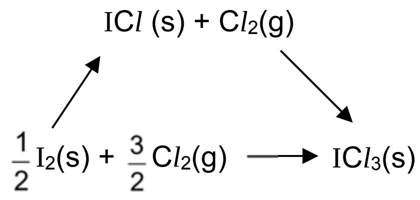
The reverse of the above reaction results in 565 kJ of energy is released.

10 Iodine trichloide, ICl_3, is made by reacting iodine with chlorine.

$$I_2(s) + Cl_2(g) \rightarrow 2ICl(s); \quad \Delta H^\theta = +14 \text{ kJ mol}^{-1}$$

$$ICl(s) + Cl_2(g) \rightarrow ICl_3(s); \quad \Delta H^\theta = -88 \text{ kJ mol}^{-1}$$

By using the above data and the following energy cycle, what is the enthalpy change of the formation of solid iodine trichloride?



A –60 kJ mol^{-1} **B** –74 kJ mol^{-1} **C** –81 kJ mol^{-1} **D** –162 kJ mol^{-1}

Answer: C

$\Delta H = +14/2 + (-88) = -81$ kJ mol^{-1}

11 The decomposition

$$2N_2O_5 \rightarrow 4NO_2 + O_2$$

is first order with respect to N_2O_5.

In an experiment 0.10 mol of pure N_2O_5 was put into an evacuated flask. It was found that there was 0.025 mol of N_2O_5 left 34 minutes later.

What is the time taken for the amount of NO_2 to rise from 0 mol to 0.10 mol?

A 17 minutes **B** 34 minutes **C** 68 minutes **D** 136 minutes

Answer: A

$$2N_2O_5 \rightarrow 4NO_2 + O_2$$

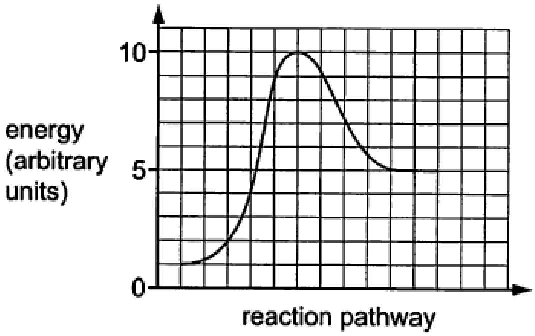
$N_2O_5 : NO_2 = 1 : 2$

For N_2O_5, 0.10 → 0.05 → 0.025 →0.0125 → 0.00625

For NO_2, 0 → 0.10 → 0.15 → 0.175 → 0.1875

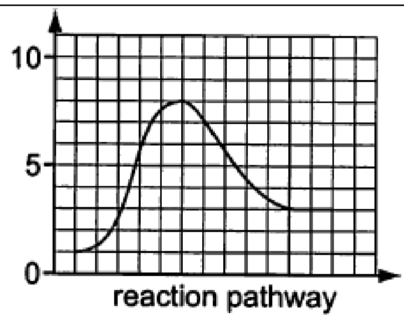
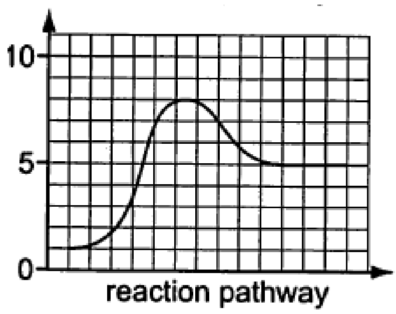
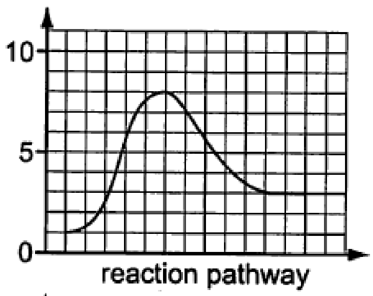
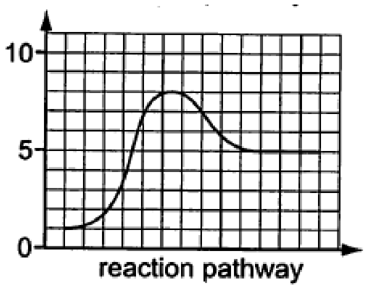
Every 1 → represents one half life (17 mins).

12 The diagram shows the reaction pathway diagram for an uncatalysed reaction.



The reaction is then catalysed.

What are the changes in the rate constant and the reaction pathway diagram?

	rate constant	energy profile
A	decrease	
B	decrease	
C	increase	
D	increase	

Answer: D

When catalyst is added, alternative pathway with lower activation energy is available. This also results in larger rate constant k. The enthalpy change of reaction remains the same.

13 Which one of the following statements about the forward and backward reactions, $P + Q \rightleftharpoons R + S$, is correct when the system is at equilibrium?

A The ratio of the rates of the forward reaction to that of the reverse reaction equals the equilibrium constant.

B The rates of both the forward and the reverse reactions are equal to zero.

C The rates of the forward and reverse reactions are equal.

D The rate constant for the forward reaction equals the rate constant for the reverse reaction.

Answer: C

Option A is wrong. The ratio of the forward rate constant to that of the reverse rate constant equals the equilibrium constant.

Option B is wrong. The rates of both the forward and the reverse reactions are equal (but not = 0).

Option D is wrong. The rate for the forward reaction equals the rate of the reverse reaction.

14 Each of the following equilibria is subjected to two changes carried out separately:

(i) the pressure is reduced at constant temperature;
(ii) the temperature is increased at constant pressure.

For which equilibrium will both of these changes result in an increase in the proportion of products?

A $H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$; $\Delta H = +53 \text{ kJ mol}^{-1}$

B $4NH_3(g) + 5O_2(g) \rightleftharpoons 4NO(g) + 6H_2O(g)$; $\Delta H = -950 \text{ kJ mol}^{-1}$

C $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$; $\Delta H = -92 \text{ kJ mol}^{-1}$

D $N_2O_4(g) \rightleftharpoons 2NO_2(g)$; $\Delta H = +57 \text{ kJ mol}^{-1}$

Answer: D

When pressure is reduced, by Le Chatelier's Principle, the equilibrium position shifts to the side that have more gaseous molecules to increase the pressure. To increase the the proportion of products when pressure is reduced, the products must have more gaseous molecules than reactants. Only Option B and D satisfy this.

When temperature is increased, by Le Chatelier's Principle, the equilibrium position shifts to the side that favors endothermic reaction to decrease tempearture. To increase the the proportion of products when temperature is increased, the forward reaction must be endothermic. Hence, Option D is answer.

15 The value of the ionic product of water, K_w, varies with temperature.

Temperature / °C	K_w / mol^2 dm^{-6}
25	1.0×10^{-14}
62	1.0×10^{-13}

What can be deduced from this information?

A The ionic dissociation of water is an exothermic process.

B The association of water molecules by hydrogen bonding increases as temperature increases.

C The pH of pure water increases with temperature.

D At 62 °C, water with a pH of 6.5 is neutral.

Answer: D

Option A is wrong as ionic dissociation of water involves bond breaking of water molecules, this is an endothermic reaction.

Option B is wrong. As temperature increases, more hydrogen bonds between molecules are broken.

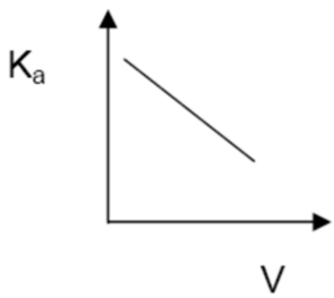
Option C is wrong. The pH of pure water decreases with temperature. (pH = 7 at 25 °C and pH = 6.5 at 62 °C)

Option D is correct as water is neutral at all temperature as $[H^+] = [OH^-]$.

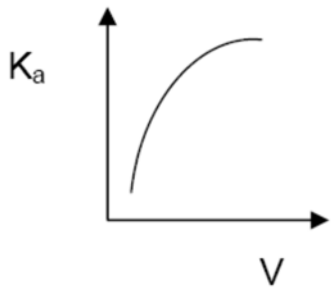
16 1 mol sample of ethanoic acid is diluted at constant temperature to a volume V.

Which diagrams shows how K_a, the acid dissociation constant, varies with V?

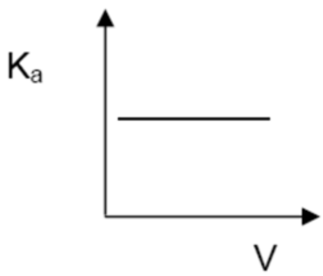
A



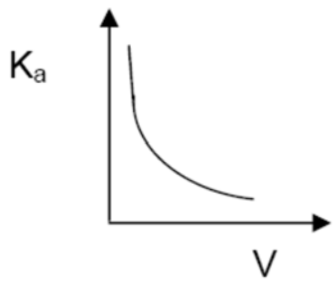
C



B



D



Answer: B Ka is temperature dependent only. (independent of volume of solution)

17 The table shows some data on two acid-base indicators.

indicator	approximate pH range	colour change	
		acid	alkali
bromocresol green	3.8 – 5.5	yellow	blue
phenol red	6.8 – 8.5	yellow	red

Which conclusion can be draw about a solution in which bromocresol green is blue and phenol red is yellow?

A It is weakly acidic.

B It is neutral.

C It is weakly alkaline.

D It is strongly alkaline.

Answer: A

When bromocresol green is added into the solution, the blue colour indicates that the pH of the solution is more than 5.5.

When phenol red is added into the solution, the yellow colour indicates that the pH of the solution is less than 6.8.

For a solution to have pH that is more than 5.5 and less than 6.8, the solution is weakly acidic.

18 Which of the following elements forms an insoluble oxide and a chloride which is readily hydrolysed?

A magnesium

B phosphorus

C silicon

D sodium

Answer: C

Option A is wrong. Magnesium oxide is slighly soluble in water and magnesium chloride undergoes hydration and partial hydrolysis.

Option B is wrong. Phosphorus pentoxide and phosphorus chloride is readily hydrolysed to form an acidic solution.

Option C is correct. Silicon dioxide is insoluble in water due to its giant covalent structure. Silicon tetrachloride is hydrolysed readily to form SiO_2 and HCl.

Option D is wrong. Sodium oxide reacts with water to form sodium hydroxide. Sodium chloride dissolves in water readily but does not undergo hydrolysis.

19 In the preparation of silicon, silicon dioxide is heated with magnesium.

$$SiO_2 + 2Mg \rightarrow 2MgO + Si$$

The product mixture contains MgO and Si only.

To separate the silicon from the product mixture, students proposed the following two possible methods.

1. Shake the mixture with aqueous hydrochloric acid and filter.
2. Heat the mixture gently and collect the evaporated silicon.

Which methods would work?

A 1 and 2 **B** 1 only **C** 2 only **D** neither 1 or 2

Answer: B

Option 1 is correct as MgO is a basic oxide that can react with HCl to undergo acid-base reaction to form salt and water. Si does not react with HCl(aq) due to its giant covalent structure.
$MgO + 2HCl \rightarrow MgCl_2 + H_2O$

Option 2 is wrong. It's almost impposible to collect evaporated silicon as it has extremely high boiling point.

20 Which statements concerning the third period elements (sodium to argon) and their compounds are correct?

1 The elements become more electronegative from sodium to chlorine.

2 The melting points of the elements decrease across the period.

3 The maximum oxidation state is shown by silicon.

4 Aluminium oxide is the only oxide which is amphoteric.

A 1 and 2 only

B 1 and 4 only

C 2 and 3 only

D 3 and 4 only

Answer: B

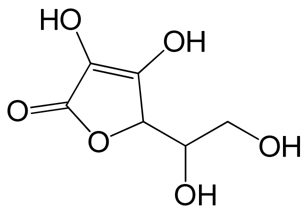
Option 1 is correct. Moving across period, nuclear charge increase, shielding effect remains relatively constant, effective nuclear charge increase. Valence electrons are more strongly attracted and electronegativity increases.

Option 2 is wrong. Melting point increases from Na to Si. Melting point of $S_8 > P_4 > Cl_2 >$Ne

Option 3 is wrong. The maximum oxidation state is shown by chlorine (+7).

Option 4 is correct. Aluminium oxide is the only oxide which is amphoteric as it can react with both acid and base.
$Al_2O_3 + 6HCl \rightarrow 2AlCl_3 + 3H_2O$
$Al_2O_3 + 2NaOH + 3H_2O \rightarrow 2NaAl(OH)_4$

21 Vitamin C has the following structure.

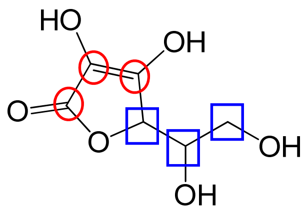


Vitamin C

How many sp^2 and sp^3 hybridised carbon atoms does this molecule have?

	sp^2	**sp^3**
A	6	0
B	4	2
C	3	3
D	2	4

Answer: C



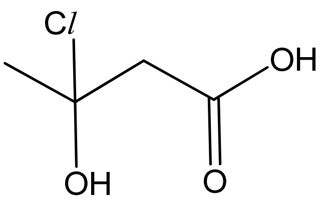
Vitamin C

sp^2 carbon: ◯

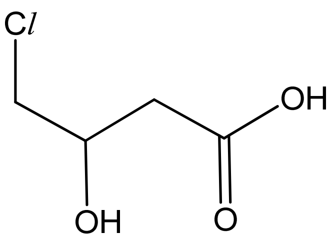
sp^3 carbon: □

22 Which structure correctly represents 4-chloro-3-hydroxybutanoic acid?

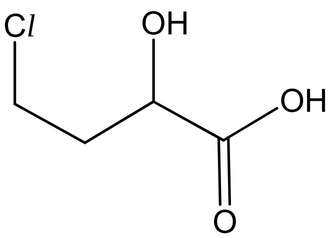
A



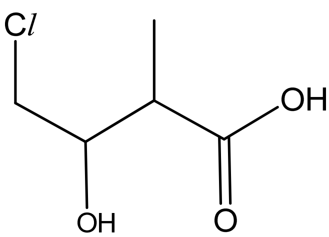
B



C



D



Answer: B

23 As the number of carbon atoms in a homologous series of alkane molecules **increases**, for which property of the alkanes does the numerical value **decrease**?

A density

B enthalpy change of vaporization

C number of isomers

D vapour pressure

Answer: D

Option A is wrong. As the number of carbon atom increases, the Mr of the molecule increases and density also increases.

Option B is wrong. As the number of carbon atom increases, the i.d.-i.d between the molecules increases as elctron cloud size increases, boiling point increases. The enthalpy change of vaporization also increases.

Option C is wrong. As the number of carbon atom increases, there are more number of isomers.

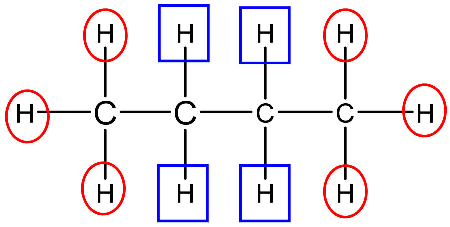
Option D is correct. As the number of carbon atom increases, the i.d.-i.d between the molecules increases as elctron cloud size increases, boiling point increases. Less molecules are vaporised and vapour pressure decreases.

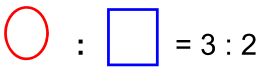
24 Compound **E**, C_4H_{10}, reacts with chlorine gas in the presence of light to form two monochlorinated alkanes, **F** and **G**, in an approximate molar ratio of 2 : 3.

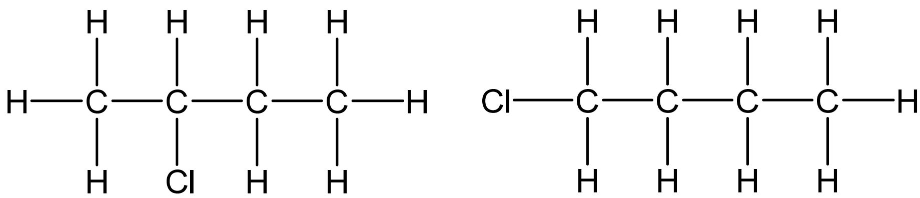
What are the structures of **E** and **F**?

	E	F
A	$CH_3CH(CH_3)CH_3$	$CH_3CH(CH_3)CH_2Cl$
B	$CH_3CH(CH_3)CH_3$	$CH_3CCl(CH_3)CH_3$
C	$CH_3CH_2CH_2CH_3$	$CH_3CH_2CHClCH_3$
D	$CH_3CH_2CH_2CH_3$	$CH_3CH_2CH_2CH_2Cl$

Answer: C







F G

25 How many different alkenes, including cis-trans isomers, could be produced when 3-methylpentan-2-ol reacts with excess concentrated sulfuric acid at 170 ^{o}C?

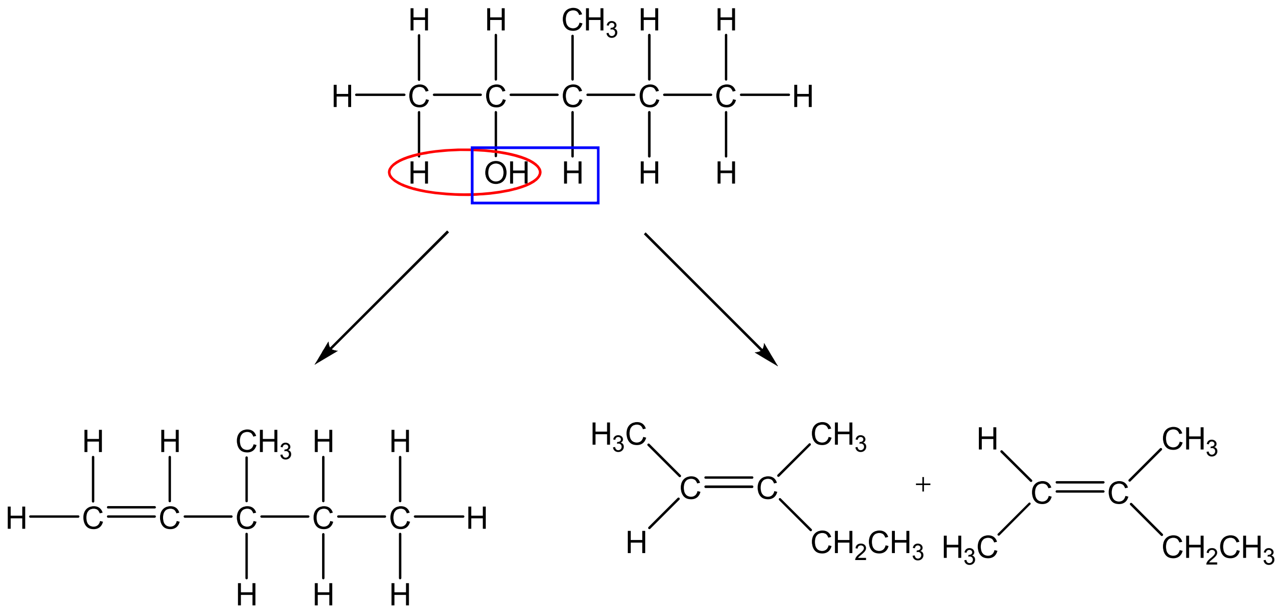
A 2

B 3

C 4

D 5

Answer: B



26 Lactic acid builds up in muscles when oxygen is in short supply. It can cause muscular pain. Part of the reaction sequence is shown.

$$\underset{\text{glyceraldehyde}}{CH_2OHCH(OH)CHO} \rightarrow \underset{\text{pyruvic acid}}{CH_3COCO_2H} \rightarrow \underset{\text{lactic acid}}{CH_3CH(OH)CO_2H}$$

Which statements about the reaction sequence are correct?

1 A secondary alcohol is oxidised to a ketone.

2 A ketone is reduced to a secondary alcohol.

3 An aldehyde is oxidised to a carboxylic acid.

4 A carboxylic acid is reduced to a primary alcohol.

A 1 and 2 only

B 1 and 4 only

C 1, 2 and 3 only

D 2, 3 and 4 only

Answer: C

Step 1: oxidation of secodary alcohol to ketone and oxidation of aldehyde to carboxylic acid

Step 2: reduction of ketone to secondary alcohol

27 A food chemist wants to create the odour of green apples for a product. An ester with this odour has the formula $CH_3CH_2CO_2CH(CH_3)_2$. In which of the following will the substances react together to produce this ester?

A CH_3CH_2OH and $(CH_3)_2CHCOOH$

B CH_3COOH and $CH_3CH(OH)CH_2CH_3$

C CH_3CH_2COOH and $CH_3CH_2CH_2OH$

D CH_3CH_2COOH and $(CH_3)_2CHOH$

Answer: D

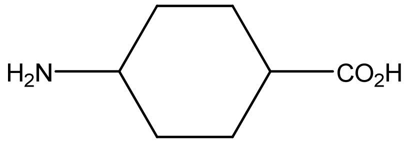
$CH_3CH_2CO_2CH(CH_3)_2 \rightarrow CH_3CH_2COOH + (CH_3)_2CHOH$ (acidic hydrolysis)

28 Which compound could be used by itself to form a condensation polymer?

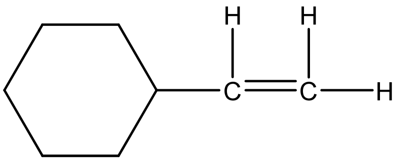
A $HOCH_2CH_2OH$

B $HO_2C(CH_2)_4CO_2H$

C



D



Answer: C

Option A cannot form condensation polymer as it only contains alcohol functional group.

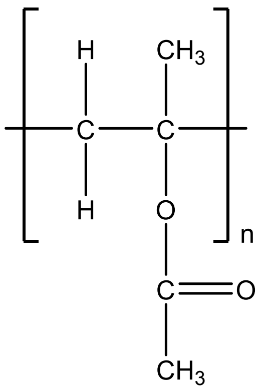
Option B cannot form condensation polymer as it only contains carboxylic acid functional group.

Option C can form condensation polymer as amine group and carboxylic acid group undergoes condensation to form amide.

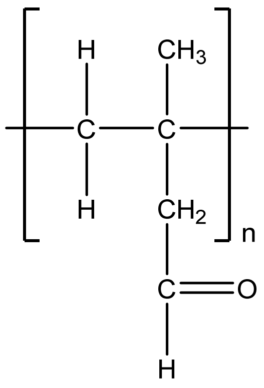
Option D cannot form condensation polymer but can form addition polymer.

29 The following polymers could be used for contact lenses. Contact lenses have to absorb water so that they can fit comfortably in the eyes. Which polymer will be the most suitable to be used for contact lens?

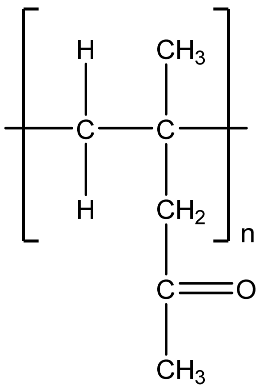
A



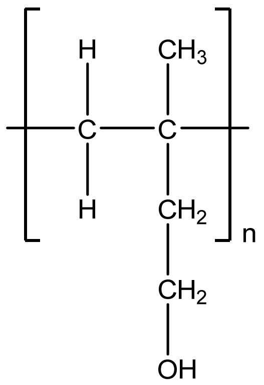
B



C



D



Answer: D

Structure D contains polar –OH group that can form hydrogen bonds with water, making it able to absorb water.

30 Which are the possible ways do nanomaterials enter the human body?

1 skin contact
2 inhalation
3 orally taken

A 1 only
B 1 and 2 only
C 2 and 3 only
D 1, 2 and 3 only

Answer: D

End of Paper